

Identification and Control of a Propeller-Actuated Damped Pendulum (PADP)

Eric Castro Velásquez*, Esteban Segura Garcia*, Daniel Chavarria Garcia*

*School of Electronics Engineering, Instituto Tecnológico de Costa Rica (ITCR), 30101 Cartago, Costa Rica
{ecastro, essegura, dachavarria}@estudiantec.cr

Abstract—This paper presents the experimental identification and closed-loop control of a propeller-actuated damped pendulum (PADP). Three step-response datasets were collected and processed using both manual parameter approximation and MATLAB's System Identification toolbox, yielding six candidate transfer functions. Statistical analysis identified one experiment as disturbance-affected and excluded it; the final plant model was obtained by averaging the four remaining models. Three continuous-time controllers were designed — a PID tuned via IMC in SISOTool, an integral state feedback regulator via LQR (ISF-LQR), and an I-PD by pole placement — and physically validated against a settling time of 5 s and maximum overshoot of 3%. The ISF-LQR and I-PD controllers satisfied both specifications, while the PID exhibited the poorest transient response among the three.

I. INTRODUCTION

The propeller-actuated damped pendulum (PADP) is an underactuated mechanical system whose angular position is governed by the aerodynamic torque of a brushless motor and propeller assembly, opposing gravity and bearing damping. Around a fixed operating point the nonlinear dynamics admit a linear second-order underdamped approximation [1], making it a compact yet representative platform for comparing identification and control strategies.

Accurate plant identification is the prerequisite for any model-based controller. Parametric methods extract the natural frequency ω_n and damping ratio ζ directly from measured transient features such as peak overshoot and oscillation period [2]. Data-driven tools, such as MATLAB's *ident* toolbox, complement this by fitting transfer function prototypes to minimize a prediction-error criterion. Because experimental data are always subject to noise and occasional disturbances, statistical screening across repeated trials is essential before any model averaging is performed.

Given a validated plant model, different control architectures offer distinct trade-offs. The PID with Internal Model Control (IMC) tuning [3] is widely used for its simplicity, but its derivative action on the error signal can amplify noise and produce overshoot in oscillatory plants. The Linear Quadratic Regulator with an augmented integral state (ISF-LQR) provides a systematic, optimization-based synthesis that simultaneously penalizes control effort and tracking error, and guarantees zero steady-state offset [4]. The I-PD structure — where only the integrator acts on the error while the proportional and derivative terms act exclusively on the plant

output — eliminates derivative kick and typically reduces overshoot compared to the parallel PID [5]; pole placement then allows the designer to prescribe closed-loop dynamics directly from the time-domain specifications.

This paper reports the complete identification-to-validation cycle for the PADP. Section II covers data acquisition, parameter estimation, statistical model selection, and the derivation of the final plant. Section III details the three controller designs. Sections IV and V present and interpret the experimental results against the design requirements of settling time $t_s \leq 5$ s and maximum overshoot $M_p \leq 3\%$.

II. SYSTEM IDENTIFICATION

A. Experimental Setup and Data Acquisition

The PADP platform consists of an extruded-aluminium support rigidly fixed to a wooden base. A hollow aluminium arm pivots on bearings at the top of the support and carries a Hobbymate 2204 brushless motor governed by a SimonK 20 A ESC fitted with a GEMFAN 5045BNR three-blade propeller. Angular position is measured by a USDigital E5 incremental encoder (3600 CPR) coupled to the pivot shaft and sampled every 20 ms with a gain $k = 2\pi/3600$ rad/count. PWM actuation uses OneShot125 mode at a 500 μ s period. Three independent experiments were conducted by applying a step input of 1.7 units from rest, establishing a steady-state operating point near $\theta_{eq} \approx 0.6$ rad ($\approx 34.4^\circ$), and recording the angular response for 40 s.

B. Parameter Approximation

The logarithmic decrement method was applied to each trial to extract ω_n , ζ , and the static gain K from the measured transient features (successive peak amplitudes and oscillation period):

$$\delta = \frac{\ln(y_0/y_i)}{i}, \quad \zeta = \frac{\delta}{\sqrt{(2\pi)^2 + \delta^2}}, \quad \omega_n = \frac{2\pi/T}{\sqrt{1 - \zeta^2}} \quad (1)$$

yielding three second-order transfer functions of the prototype:

$$G_i(s) = K_i \frac{\omega_{n,i}^2}{s^2 + 2\zeta_i \omega_{n,i} s + \omega_{n,i}^2}, \quad i = 1, 2, 3 \quad (2)$$

TABLE I
PARAMETERS ESTIMATED BY THE LOGARITHMIC DECREMENT METHOD

Trial	K	ζ	ω_n (rad/s)
1	0.338	0.032	3.9292
2	0.340	0.035	3.9294
3	0.423	0.033	3.9291

C. MATLAB System Identification

MATLAB's *ident* toolbox was used in parallel to fit the same second-order prototype to each dataset via prediction-error minimization, producing three additional transfer functions with their respective best-fit percentages.

TABLE II
PARAMETERS ESTIMATED BY SYSTEM IDENTIFICATION

Trial	K	ζ	ω_n (rad/s)
1	0.346	0.0292	3.9225
2	0.346	0.0295	3.9301
3	0.426	0.0296	3.9091

D. Statistical Analysis and Model Selection

The six identified models were pooled and their parameters compared statistically. The mean $\bar{x}$ and sample standard deviation s were computed for K , ζ , and ω_n . The two models derived from trial 3 showed parameter deviations exceeding $2s$ from the group mean, consistent with an external disturbance during that recording session. Those two models were therefore excluded, and the final plant was obtained by averaging the parameters of the four remaining transfer functions:

$$G(s) = \frac{K_{\text{avg}} \omega_{n,\text{avg}}^2}{s^2 + 2 \zeta_{\text{avg}} \omega_{n,\text{avg}} s + \omega_{n,\text{avg}}^2} \quad (3)$$

$$G(s) = \frac{5.287}{s^2 + 0.2481s + 15.43} \quad (4)$$

III. CONTROLLER DESIGN

Design specifications for all regulators: settling time $t_s \leq 5$ s (2% criterion) and maximum overshoot $M_p \leq 3\%$ to a 0.6 rad step reference, with zero steady-state error and disturbance rejection.

A. Continuous PID Controller (SISOtool)

A continuous-time PID controller was designed using MATLAB's SISOtool with automatic tuning, based on the identified plant model.

$$C_{\text{PID}}(s) = K_p + \frac{K_i}{s} + K_d s \quad (5)$$

B. Integral State Feedback Controller (ISF) via LQR

An Integral State Feedback (ISF) controller was designed using the Linear Quadratic Regulator (LQR) criterion. The method augments the plant with an integral state to ensure zero steady-state error and improve disturbance rejection.

The identified plant in controllable canonical form is:

$$\mathbf{A} = \begin{bmatrix} -0.2481 & -15.4300 \\ 1 & 0 \end{bmatrix}, \quad \mathbf{B} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad \mathbf{C} = [0 \quad 5.2870], \quad \mathbf{D} = 0 \quad (6)$$

The system is fully controllable (rank = 2). To include integral action, the augmented model is defined as:

$$\dot{x}_I = r - \mathbf{C}\mathbf{x} \quad (7)$$

$$\mathbf{A}_a = \begin{bmatrix} -0.2481 & -15.4300 & 0 \\ 1 & 0 & 0 \\ 0 & -5.2870 & 0 \end{bmatrix}, \quad \mathbf{B}_a = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \quad (8)$$

The augmented system is also controllable (rank = 3). Based on the specifications ($t_s \leq 5$ s, $M_p \leq 3\%$), the desired poles were selected as:

$$p_{1,2} = -2.8 \pm 3.1266j, \quad p_3 = -14.0 \quad (9)$$

The LQR weighting matrices were chosen using Bryson's rule:

$$\mathbf{Q} = \text{diag}\left(\frac{1}{9^2}, \frac{1}{7^2}, \frac{1}{0.45^2}\right), \quad R = \frac{1}{0.6^2} \quad (10)$$

The resulting gain vector $\hat{\mathbf{K}} = [k_1 \quad k_2 \quad K_i]$ is:

$$\hat{\mathbf{K}} = [0.7214 \quad 0.4370 \quad -1.3333] \quad (11)$$

which yields the closed-loop poles:

$$p_{1,2} = -0.2593 \pm 3.9454j, \quad p_3 = -0.4509 \quad (12)$$

C. I-PD Controller via Pole Placement

The I-PD structure places the integrator on the error signal while the proportional and derivative terms act on the plant output, preventing derivative kick on step changes of the reference:

$$C_{\text{I-PD}}(s) = \frac{K_i}{s} - (K_p + K_d s) \quad (13)$$

The target specifications ($t_s \leq 5$ s, $M_p \leq 3\%$) were mapped to the damping ratio ζ_d and natural frequency $\omega_{n,d}$ of the desired dominant poles. A third real pole was placed sufficiently far to the left to remain non-dominant. Coefficient matching of the closed-loop characteristic polynomial yielded K_p , K_d , and K_i , subsequently decomposed into P, I, D, N parameters for the PASCAL PID block. The controller was designed to specifications more strict than the nominal requirements ($t_s \leq 2.9$ s, $M_p \leq 1\%$) to provide a performance margin that compensates for the degradation expected during physical implementation.

$$p_{1,2} = -1.3793 \pm j0.4705, \quad p_3 = -6.8966$$

$$K_p = 1.0817, \quad K_i = 2.7704, \quad K_d = 1.7793, \quad N = 20$$

The third real pole p_3 was placed at five times the magnitude of the real part of the dominant poles, following the standard non-dominance criterion [1]:

$$|p_3| = 5 |\text{Re}(p_{1,2})| = 5 \times 1.3793 = 6.897$$

This placement ensures that the transient contribution of p_3 decays approximately five times faster than the dominant pair

D. Discrete PID Controller (SISObtool)

A discrete-time PID controller was designed using SISObtool with a sampling period of $T = \dots$ s, employing the ... discretization method.

IV. RESULTS AND ANALYSIS

This section presents the results obtained in both simulation and physical implementation for each controller, evaluating compliance with the performance specifications ($t_s \leq 5$ s, $M_p \leq 3\%$).

A. Continuous PID Controller

B. Integral State Feedback Controller via LQR

The closed-loop step responses of the ISF-LQR controller for $r = 0.6$ rad are shown in Fig. 1 (experiment) and Fig. 2 (simulation). The corresponding performance metrics are summarized in Table III.

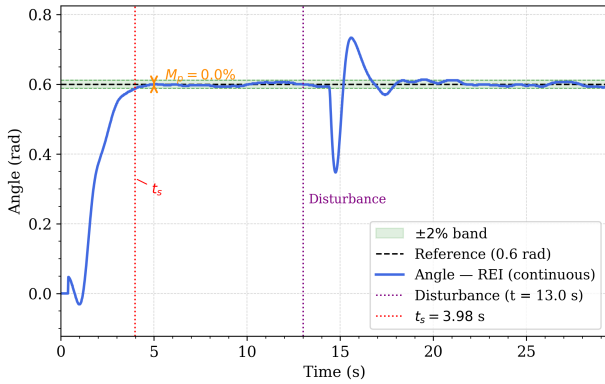


Fig. 1. Experimental step response of the ISF-LQR controller.

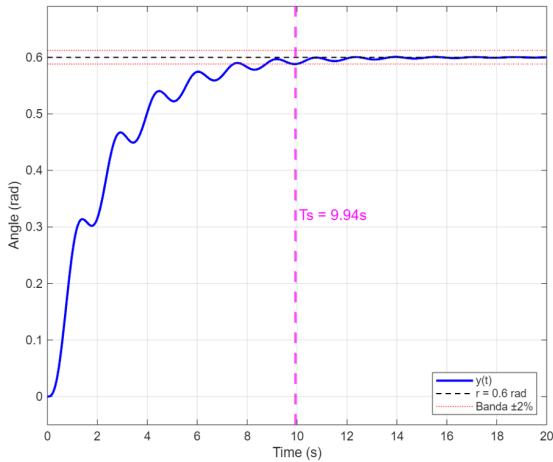


Fig. 2. Simulated step response of the ISF-LQR controller.

TABLE III

ISF-LQR PERFORMANCE METRICS: SIMULATION VS. EXPERIMENT

Condition	t_s (s)	M_p (%)	e_{ss}	Meets specs?
Simulation	9.94	0.12	0	No
Experimental	3.98	0	0	Yes

As observed, the simulated response exhibits a settling time of 9.94 s, exceeding the design requirement of 5 s, although the overshoot remains negligible at 0.12 %. In contrast, the experimental implementation achieves a significantly faster settling time of 3.98 s with no observable overshoot, fully complying with the specified performance criteria. Both cases satisfy the steady-state error requirement ($e_{ss} = 0$) due to the inclusion of integral action. Additionally, the overshoot constraint ($M_p \leq 3\%$) is met in both simulation and experimental results.

C. I-PD Controller via Pole Placement

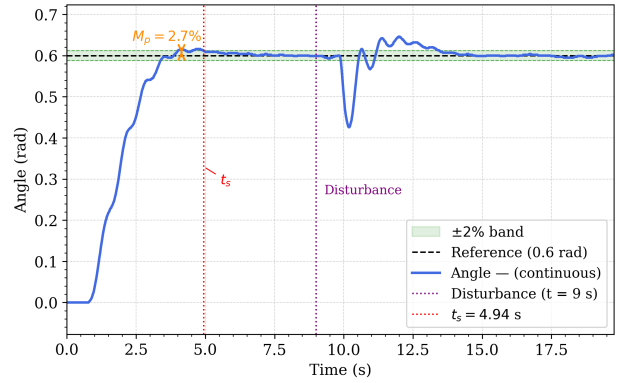


Fig. 3. Experimental step response of the ISF-LQR controller.

D. Discrete PID Controller

E. Comparative Summary

V. DISCUSSION

This section interprets the results presented above, analyzing the causes of the observed differences between theoretical models and the physical behavior of the system, as well as the advantages and disadvantages of each implemented control strategy.

A. ISF-LQR: Simulation–Experiment Discrepancy

Figs. 2 and 3 show a clear contrast between simulated and experimental responses. The simulation exhibits a slowly damped transient with $t_s = 9.94$ s, consistent with the real pole $p_3 = -0.4509$ near the imaginary axis, which dominates the settling behavior despite the faster complex poles. In contrast, the experimental response converges monotonically to the 0.6 rad reference in $t_s = 3.98$ s with zero overshoot, satisfying both specifications.

A small negative deflection at $t \approx 1$ s in the experimental response (about -0.07 rad) is attributed to actuator dead-zone

and propeller inertia, which delay the onset of effective control action. This effect is not captured by the linear model.

The faster experimental convergence is consistent with additional nonlinear damping (e.g., friction and aerodynamic drag), which increases the effective damping and suppresses oscillations, leading to a shorter settling time than predicted.

Finally, the experimental response shows effective disturbance rejection; a perturbation at $t \approx 13$ s produces a deviation of approximately 0.25 rad, after which the output returns to the reference, remaining within the margin of less than 2% steady-state error. This confirms the role of comprehensive action in rejecting disturbances despite the model's inaccuracies.

VI. CONCLUSIONS

The ISF-LQR controller achieved zero steady-state error and effective disturbance rejection. While the simulation resulted in $t_s = 9.94$ s and $M_p = 0.12\%$, failing the settling time requirement, the experimental response reached $t_s = 3.98$ s with $M_p = 0\%$, satisfying all specifications.

REFERENCES

- [1] K. Ogata, *Modern Control Engineering*, 5th ed. Prentice Hall, 2010.
- [2] L. Ljung, *System Identification: Theory for the User*, 2nd ed. Prentice Hall, 1999.
- [3] K. J. Åström and T. Hägglund, *Advanced PID Control*. ISA, 2006.
- [4] B. D. O. Anderson and J. B. Moore, *Optimal Control: Linear Quadratic Methods*. Prentice Hall, 1990.
- [5] A. Visioli, *Practical PID Control*. Springer, 2006.
- [6] G. F. Franklin, J. D. Powell, and M. L. Emami-Naeini, *Feedback Control of Dynamic Systems*, 4th ed. Prentice Hall, 1998.